

P1) $Y_1, \dots, Y_n \stackrel{iid}{\sim} B_1(m, p)$ $p \in [0, 1]$, $m \in \mathbb{N}$ conocido

$$P(x) = \binom{m}{x} p^x (1-p)^{m-x}$$

$$f(\vec{Y} | m, p) = \prod_{i=1}^n \binom{m}{Y_i} p^{Y_i} (1-p)^{m-Y_i}$$

$$= \prod_{i=1}^n \binom{m}{Y_i} \left[p^{\sum_{i=1}^n Y_i} (1-p)^{n \cdot m - \sum_{i=1}^n Y_i} \right]$$

$$= \vec{Y} \in \{0, 1, \dots, m\}^n$$

$$p \in [0, 1], \chi$$

Obs: la cantidad de
muestras n es
igual al parámetro
de la binomial.
Caso especial

P2) $Y_i | x_i \stackrel{iid}{\sim} N(m(x_i), \sigma^2)$ $i=1 \dots n$ $x_i \in \mathbb{N}$ conocido

$$f(Y_i | x_i, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(Y_i - m(x_i))^2}{2\sigma^2}}$$

$$f(\vec{Y} | \vec{x}, \sigma^2) = 2\pi^{-n/2} \sigma^{-n} e^{-\frac{\sum_{i=1}^n (Y_i - m(x_i))^2}{2\sigma^2}}$$

$$\vec{Y} \in \mathbb{N}^n$$

$$\sigma \in \tilde{\mathbb{T}} \times \mathbb{N}^+$$

↓

familia
de funciones
continuas

$|\tilde{\mathbb{T}}| > \mathbb{N} \rightarrow$ no paramétrica

P2) $Y_i \sim \text{Bin}(n=1000, p) \quad i=1, \dots, 50$

$$P(\vec{Y} | n, p) = \prod_{i=1}^{50} \binom{n}{y_i} p^{y_i} (1-p)^{n-y_i}$$

$$= \prod_{i=1}^{50} \binom{n}{y_i} \left\{ p^{y_i} (1-p)^{n-y_i} \right\}$$

$$X = \{1, \dots, 1000\}^{50}$$

$$\theta \in [0, 1]$$

$$Y = \sum_{i=1}^n Y_i \sim \text{Bin}(m=50 \cdot 1000, p)$$

$$p(X) = \binom{m}{y} p^y (1-p)^{m-y}$$

$X \in Y$ independent

$$X \sim \text{Bin}(m_x, p) \\ Y \sim \text{Bin}(m_y, p)$$

$$Z = X + Y$$

$$p(Z=z) = P(X=z, Y=0)$$

$$= \sum_{y=0}^z P(X=z-y, Y=y) P(Y=y)$$

$$= \sum_{y=0}^z \binom{m_x}{z-y} p^{z-y} (1-p)^{m_x-z+y} \binom{m_y}{y} p^y (1-p)^{m_y-y}$$

$$p^y$$

$$(1-p)^{m_y-y}$$

$$p^{z-y}$$

$$(1-p)^{m_x-z+y}$$

$$= \sum_{y=0}^z \binom{m_x}{z-y} \binom{m_y}{y} \frac{m_x!}{(z-y)! (m_x-z+y)!} \frac{m_y!}{y! (m_y-y)!} p^z (1-p)^{m_x+m_y-z}$$

$$M_X = (1-p) + p e^t$$

$$M_Y = (1-p) + p e^t$$

$$Z \sim \text{Bin}(m_x + m_y, p)$$

P3) Identifizierbarkeit

$$F(x, \sigma_1) = F(x, \sigma_2) \rightarrow \sigma_1 = \sigma_2 \quad \forall x$$

$$\forall \sigma_1, \sigma_2 \in \mathcal{O}$$

$$1) Y_i = a + b_1 x_i + b_2 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

$$Y_i \sim N(a + b_1 x_i + b_2 x_i, \sigma^2)$$

$$f(\vec{y} | a, b_1, b_2, x_i, \sigma^2) = \frac{(2\pi)^{-n/2} \sigma^{-n} \exp\left\{-\frac{1}{2\sigma^2} \sum_i (y_i - (a + b_1 x_i + b_2 x_i))^2\right\}}{(2\sigma^2)^n}$$

$$x_i=1 \quad b_1=1 \quad x_i=1 \quad b_1=1/2 \\ b_2=1 \quad b_2=2$$

$\rightarrow$ nicht identifizierbar

$x_i=0 \rightarrow$ 1. Modell ist identifizierbar $\rightarrow \sigma^2$

wie sind Beispiele

$$2) X_1, \dots, X_n \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2)$$

$$f(\vec{x} | \mu, \sigma^2) = \frac{(2\pi)^{-n/2} \sigma^{-n} \exp\left\{-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2\right\}}{(2\sigma^2)^n}$$

$$\mu_1, \mu_2 \in \mathbb{R}, \sigma_1^2, \sigma_2^2 \in \mathbb{R}^+$$

$$f(\bar{X} | \mu_1, \sigma_1^2) = f(\bar{X} | \mu_2, \sigma_2^2)$$

$$(-2\pi)^{-n/2} \sigma_1^{-n} \exp\left\{-\frac{1}{2\sigma_1^2} \sum (y_i - \mu_1)^2\right\} = (-2\pi)^{-n/2} \sigma_2^{-n} \exp\left\{-\frac{1}{2\sigma_2^2} \sum (y_i - \mu_2)^2\right\}$$

$$e^{\log \sigma_1^{-n} \sigma_2^n} = \exp\left\{-\frac{1}{2\sigma_2^2} \sum y_i^2 - 2\mu_2 \sum y_i + n\mu_2\right\} + \frac{1}{2\sigma_1^2} \left\{\sum y_i^2 - 2\mu_1 \sum y_i + n\mu_1\right\}$$

$$e^{\log\left(\frac{\sigma_2^n}{\sigma_1^n}\right)} = \exp\left\{\sum y_i^2 \left(\frac{1}{2\sigma_2^2} - \frac{1}{2\sigma_1^2}\right) - \sum y_i \left(\frac{\mu_2}{\sigma_2^2} - \frac{\mu_1}{\sigma_1^2}\right) - \frac{n}{2} \left(\frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2}\right)\right\}$$

$$e^{n \log\left(\frac{\sigma_2}{\sigma_1}\right)}$$

$$1 = \exp\left\{\sum y_i^2 \left(\frac{1}{2\sigma_2^2} - \frac{1}{2\sigma_1^2}\right) - \sum y_i \left[\frac{\mu_2}{\sigma_2^2} - \frac{\mu_1}{\sigma_1^2}\right] - \frac{n}{2} \left[\frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2} - 2 \log\left(\frac{\sigma_2}{\sigma_1}\right)\right]\right\}$$

$$(1) \frac{1}{2\sigma_2^2} - \frac{1}{2\sigma_1^2} = 0$$

$$\downarrow$$

$$\sigma_2^2 = \sigma_1^2$$

$$(2) \frac{\mu_2}{\sigma_2^2} - \frac{\mu_1}{\sigma_1^2} = 0$$

$$\mu_2 - \mu_1 = 0$$

$$\downarrow$$

$$\mu_2 = \mu_1$$

$$(3) \frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2} - 2 \log\left(\frac{\sigma_2}{\sigma_1}\right) = 0$$

$\downarrow$
 $\text{LHS}(3)$

$$0 - 2 \log\left(\frac{\sigma_2}{\sigma_1}\right) = 0$$

$$\frac{\sigma_2}{\sigma_1} = 1 \Rightarrow \sigma_2 = \sigma_1$$

14) $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} f(x|\alpha, \beta) = \frac{2\beta^2}{\alpha^2} x \mathbb{1}_{(0, \alpha/\beta)}(x) \quad 0 < x < \frac{\alpha}{\beta}$

$f(\vec{x}) = \prod_{i=1}^n \frac{2\beta^2}{\alpha^2} x_i \mathbb{1}_{(0, \alpha/\beta)}(x_i) \quad \chi = (\alpha, \beta) \in \mathbb{R}^2 \quad \alpha, \beta > 0$

a) $= 2^n \left(\frac{\beta}{\alpha}\right)^{2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \alpha/\beta)}(x_i)$

b) $\beta = 2 \rightarrow$ da la minima distanza $\rightarrow$ no es. distrib. cond.

c) $\sigma = 2/\beta \quad \sigma > 0$

$f(\vec{x}|\sigma) = \prod_{i=1}^n 2\sigma^2 x_i \mathbb{1}_{(0, \sigma)}(x_i)$
 $= 2^n \sigma^{2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \sigma)}(x_i)$

Ass. $\sigma_1^{2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \sigma_1)}(x_i) = \sigma_2^{2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \sigma_2)}(x_i)$

$\left(\frac{\sigma_1}{\sigma_2} \right)^{2n} \prod_{i=1}^n \frac{x_i}{1} = 1 \rightarrow \sigma_1 = \sigma_2$

15) $X_1, \dots, X_n \sim \text{Poisson}(\theta) \rightarrow T(X_1, \dots, X_n) = \sum x_i$

$Z = X_1 + X_2 \rightarrow P_r(Z=z) = P_r(X_1 + X_2 = z) =$

$= P_r(X_1 = z - X_2)$

$= \sum_{x_2=0}^z P_r(X_1 = z - x_2 | X_2 = x_2) P_r(X_2 = x_2)$

$= \sum_{x_2=0}^z \frac{e^{-\theta} \theta^{z-x_2}}{(z-x_2)!} \frac{e^{-\theta} \theta^{x_2}}{x_2!} = \sum_{x_2=0}^z \frac{z!}{z-x_2! x_2!} \frac{\theta^z e^{-2\theta}}{z!}$

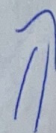
Teorema del Bin.

$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

Considera:

$2^n = \sum_{k=0}^n \binom{n}{k}$

$Z \sim \text{Poisson}(2\theta)$



Logo $T \sim \text{Poisson}(n\theta)$

$$T = \sum_{i=1}^n x_i$$

$$\frac{f(\vec{x}|\theta)}{f_T(T|\theta)} \rightarrow \text{no depende de } \theta$$

$$\frac{\prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}}{\frac{e^{-n\theta} (n\theta)^T}{T!}} = \frac{\frac{e^{-n\theta} \theta^{\sum x_i}}{\prod x_i!}}{\frac{e^{-n\theta} n^T \theta^T}{T!}} = \frac{T!}{n^T \prod_{i=1}^n x_i!} \rightarrow \text{no depende de } \theta$$

T contiene toda la información de θ que está en la muestra